Sergio **Manthey Corchado**

Particle Physicist · Ph.D. Candidate

CIEMAT / Universidad Autónoma de Madrid

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Academic Education

Universidad Autónoma de Madrid — *Madrid, Spain*

Ph.D. in Physics — *2021 - 2026 (defence pending)*

Thesis: Liquid Argon Detector Development for the Neutrino Oscillation Measurement with the DUNE experiment.

Università di Roma La Sapienza — *Rome, Italy*

M.S. in Particle and Astroparticle Physics — *2018 - 2021 (graduated 19 Jul 2021)*

Thesis: BULLKID. Low Mass Dark Matter Research with Microwave Kinetic Inductance Detectors (MKIDS).

University of Bremen — *Bremen, Germany*

B.S. in Physics — *2014 - 2018 (graduated 3 Apr 2018)*

Thesis: Oxygen-Radical Production through VUV-Light for the Surface Treatment of Adhesive and Lacquer Processes.

Work Experience

Responsibilities

- Lead analyst of the CIEMAT apparatus for the VD **X-ARAPUCA Characterisation**.
- Measurement of the absolute **Photon Detection Efficiency** of full-scale X-ARAPUCAs for the DUNE Far Detector Vertical Drift.
- Cryogenic characterisation of **SiPM** arrays – FBK NUV-HD-Cryo and Hamamatsu VUV4 – for the DUNE **Photon Detection System**.
- Software implementation of **Solar Neutrino Reconstruction** chain in LArSoft.
- Development of waveform filtering and deconvolution, photon pulse reconstruction and **charge–light matching** for the DUNE **Monte–Carlo** workflow.
- **Solar neutrino sensitivity** analyses: day–night asymmetry, HEP neutrino discovery potential and DUNE's reach on the solar oscillation parameters.
- Presentation of results at DUNE collaboration meetings and international conferences.

Positions

Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas — *Madrid, Spain*

Scientific Researcher — *2021 - 2025*

- Join professional membership of **DUNE**, **ProtoDUNE** & **SoLAr** experiments.
- Perform laboratory measurements of cryogenic **Liquid Argon Scintillation** light detection with photosensors.
- Develop **Monte-Carlo** simulations and reconstruction tools for **Low-Energy Neutrinos** analyses.
- Produce low-energy reconstruction and sensitivity analysis of **Solar Neutrinos**.
- Participate in **Outreach Program** for national secondary school.

Duke University — *Durham, USA*

Visiting researcher at Duke University — *Jul 2023 - Sep 2023*

Develop **Low-Energy Analysis Tools** in LArSoft for solar neutrino detection.

European Space Agency (ESA-ESAC) — *Villafranca del Castillo, Spain*

Visiting research contract from ESA — *Jul 2021 - Sep 2021*

Outreach for scientific project (CesaR).

Ministry of Foreign Affairs and International Cooperation (MAECI) — *Rome, Italy*

Research Under Academic Supervision (Grant) — *Jan 2020 - Jun 2020*

Detector science in the cryogenic laboratory for **Dark Matter R&D**.

Fraunhofer Institute (IFAM) — *Bremen, Germany*

Student Researcher — *Nov 2016 - May 2018*

Laboratory assistant in the department for **Adhesive Bonding and Plasma Technologies**.

Skills

Technical

Programming C++, Python, Geant4, LArSoft and LaTeX
Computing Linux, Git and HTCondor

Language

Spanish Mother tongue. Speaking: native, Reading: native, Writing: native and Listening: native
English Cambridge – C2. Speaking: proficient, Reading: proficient, Writing: proficient and Listening: proficient
German EOI Palma – C1. Speaking: advanced, Reading: advanced, Writing: advanced and Listening: advanced
Catalan C1. Speaking: advanced, Reading: advanced, Writing: advanced and Listening: advanced
Italian Bremen University – A2. Speaking: intermediate, Reading: intermediate, Writing: intermediate and Listening: intermediate

Conferences

Lake Louise Winter Institute 2025 *Alberta, Canada*
Talk *02/03/2025 - 08/03/2025*
Low energy and BSM searches in DUNE.

Light Detection In Noble Elements Conference 2024 *Sao Paulo, Brazil*
Talk *26/08/2024 - 28/08/2024*
X-ARAPUCA Photon Detection Efficiency for DUNE Far-Detector Vertical-Drift.

42nd International Conference on High Energy Physics *Prague, Czech Republic*
Poster *17/07/2024 - 24/07/2024*
DUNE's Low Energy Physics Searches.

XXXI International Conference on Neutrino Physics and Astrophysics *Milan, Italy*
Poster *17/06/2024 - 21/06/2024*
DUNE Sensitivity to Solar Neutrinos.

19th International Workshop on Low Temperature Detectors *Virtual*
Poster *19/07/2021 - 30/07/2021*
Status of BULLKID, an array of kinetic inductance detectors to search for Dark Matter.

Schools

Dark Matter and Neutrinos 2025 *Paris, France*
Talk *05/05/2025 - 09/05/2025*
DUNE's Sensitivity to Solar Neutrinos.

Research Activities

Deep Underground Neutrino Experiment

Madrid, Spain

PH.D. RESEARCH

2021 - 2025

- My research focuses on advancing the physics capabilities of the **Deep Underground Neutrino Experiment** (DUNE) through both hardware development and data analysis, with a particular emphasis on improving low-energy neutrino detection. My work experience has led me to perform cryogenic laboratory measurements on photosensitive devices, conduct Monte Carlo simulations, and develop reconstruction algorithms for solar neutrinos.

- A central component of my research has been the characterisation and optimisation of **X-ARAPUCA** photon detectors. These devices are a critical part of DUNE's **Photon Detection System** (PDS), responsible for capturing scintillation light produced in DUNE. Because the performance of the PDS directly affects event reconstruction, especially for low-energy interactions such as solar neutrinos, it is essential to characterise its behaviour. My work involves operating these devices in liquid nitrogen or liquid argon environments, triggering on specific events, calibrating a light source and quantifying the system's response. Through these studies, I have computed the expected **Photon Detection Efficiency** for full-scale DUNE X-ARAPUCAs and provided essential input for simulation tuning and detector optimisation. My results have been published in arXiv:2511.12328.

- Bridging the hardware and physics-analysis components of my work, I have contributed extensively to DUNE's **Monte Carlo** simulation framework, focusing on the reconstruction of low-energy charge and scintillation signals. My contributions include implementing waveform filtering and deconvolution techniques, photon pulse reconstruction, event clustering and charge-light matching for further downstream physics analyses. These efforts have been incorporated into the broader **LArSoft Simulation** workflow. A major scientific goal is the reconstruction and study of low-energy neutrinos from the Sun. DUNE's unprecedented combination of mass and high cross-section will allow it to perform precision measurements of the solar neutrino flux and probe matter effects. My research contributes to this effort through the development of vertex and energy reconstruction algorithms that combine scintillation and ionisation signals, and studying event-topology features that distinguish low-energy neutrino interactions from radiological backgrounds.

- Using these tools, I have performed analyses aimed at quantifying DUNE's solar neutrino significance, such as the **Day-Night Asymmetry**, the **HEP Neutrino Discovery** and DUNE's **Sensitivity to Solar Oscillation** parameters. My work also explores how improved photon detection translates into better energy resolution and ultimately stronger physics reach. Results from this work have been shared in doi:10.22323/1.476.0217.

- In summary, my research integrates detector development, detailed simulation, and physics analysis to enhance DUNE's ability to observe and characterise low-energy neutrinos. By improving the performance and understanding of the photon detection system, refining simulation tools, and advancing reconstruction and oscillation analysis techniques, I aim to support DUNE's long-term goal of delivering transformative neutrino physics.

BullKid

Rome, Italy

MASTER RESEARCH

2020 - 2021

- I contributed to the **BULky and Low-Threshold Kinetic Inductance Detectors** (BULLKID) R&D project, focused on the development of cryogenic **Kinetic Inductance Detectors** (KIDs) for the search of rare low-energy processes such as **Low-Mass Dark Matter** and neutrino coherent scattering. I participated in the lithographic design of these MKIDs, tuning their resonance frequencies to achieve multiplexing capabilities. I also performed data quality and noise performance studies on one of the project's first functional prototypes. I produced exclusion curves for low-mass dark matter that could potentially be achieved with a scaled version of this prototype. This work was recognised through my poster presentation at the **Low Temperature Detectors** conference.